

Lab 4: Database Administration and Users' Privileges

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1 Objectives

The purpose of this lab is to discover Database Administration through the creation of a simple sample Database on MariaDB (within Docker).

2 Reading

To be able to actively participate in this lab the student is expected to have read the lectures (obviously) and participated to the workshops (not mandatory but a very good idea). Additional reading from provided resources is beneficial but not crucial to the completion of this lab.

3 Setup

For this lab, you can use your connection to MariaDB (running with Docker on your local computer, see lecture 00 at the end).

4 Presentation

The presentation for Lab 4 is done orally. **All group members** present to the lab assistant which then determines if the demonstrated knowledge is sufficient for a pass. If not, the students can present at a later occasion, same or other day. The lab is graded individually, meaning that one team member can pass while the other fails.

5 Instructions

You **MUST** create a database "futsal_club" with 5 tables following the conceptual diagram (Figure 1):

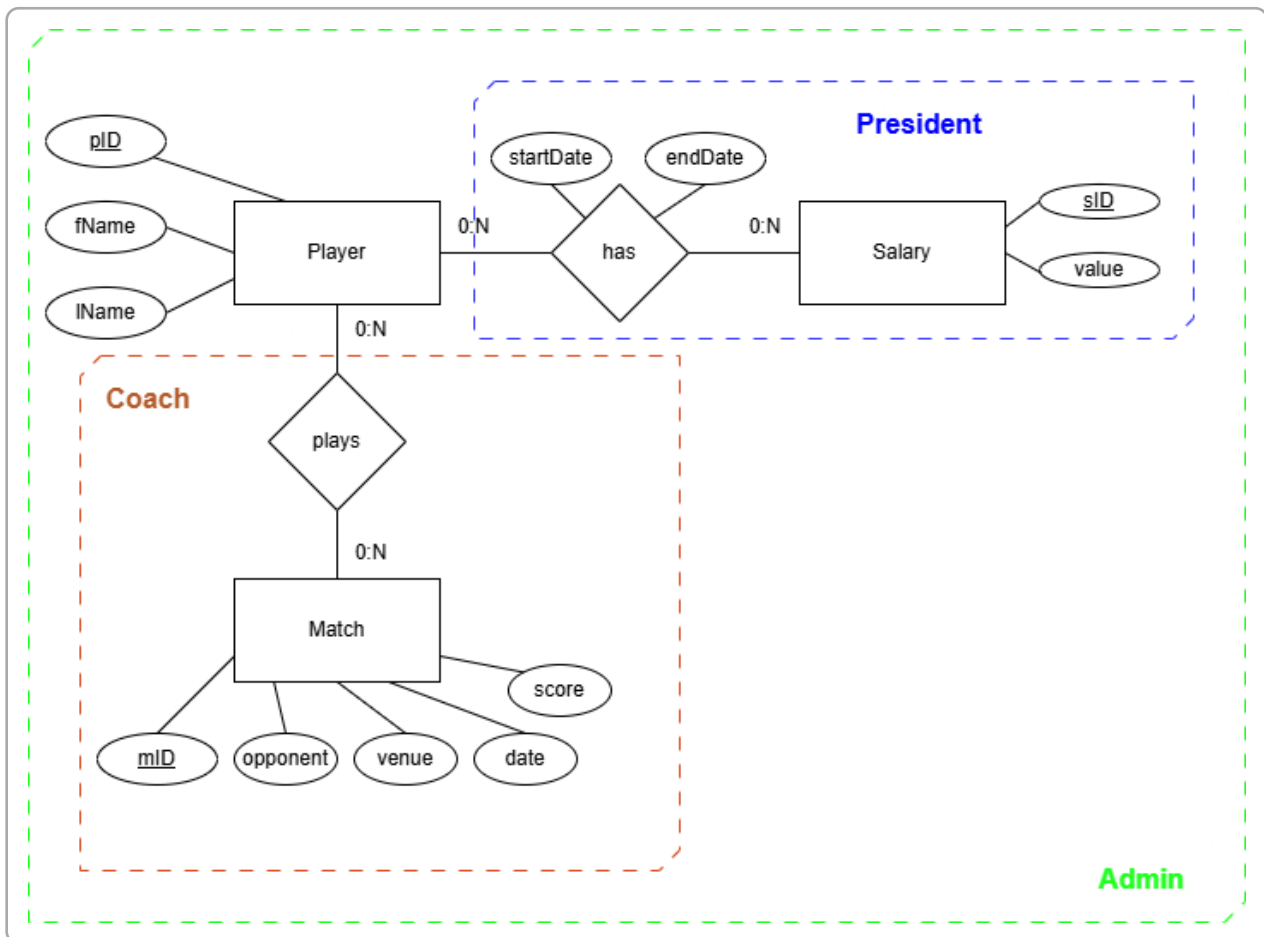


Figure 1: Entity-Relationship Diagram (ERD) for the futsal club database with users and their privileges on tables.

- Populate the table according to the constraints:
 - Player: 8 players (with realistic data)
 - Salary: 9 salaries with a player having two different salaries at two different moments in time
 - Match: 2 matches with 5 players in the team for each match
- Create THREE users as the roles in your database:
 - Admin with password "admin123" and ALL PRIVILEGES on all tables (the Admin can do anything on any table)
 - Coach with password "coach123" and ALL PRIVILEGES on "Match" and "plays" tables (the Coach can decide who plays a match and change match and plays information), READ PRIVILEGES on Player, NO PRIVILEGES at all on "Salary" and "has" tables
 - President with password "president123" and ALL PRIVILEGES on "Salary" and "has", READ PRIVILEGES on "Player" and NO PRIVILEGES at all on "plays" and "Match"
- Once everything is created, log in into MariaDB as the three users/roles you defined and TEST READ, WRITE on every table to check if your PRIVILEGES work correctly for each table and each role.
- The lab assistant will ask you to demonstrate on some tables with all three users!

PRESENT the lab to the lab assistant!!!

At the end of the lab, create a text file "lab4-yournames.sql" and put it on Canvas. This text file MUST contain ALL the SQL code to create the database, create the tables and populate them, create the users, and define their permissions (privileges). Running this file on my computer, I should be able to recreate your working environment and check the permissions of each user on my own MariaDB database!